



SUMMARY

Senior AI/ML Engineer with 9.5+ Years of Experience & Senior Data Scientist Developer 6.5+ Years of Experience in Analysis, Visualisation, Statistics, Design, Training, Testing, Deployment, CI-CD Pipeline, DS, & Implementation of various applications.

Proficient in Object-Oriented, Reusable Code, Privacy & Security, Memory Analysis. Possesses diverse experience in planning & executing multiple projects and liaising with the key stakeholders to identify & resolve business problem statement & deliver excellent results.

KEY SKILLS

• Machine/Deep Learning

Methodologies • Optimization

Techniques • Text Mining • Data

Mining & Analytics • Predictive &

Statistical Modelling • Sentiment

Analysis • Anomaly Detection • Data

Analysis • Data Visualization •

Project Delivery • Predictive

Modelling & Analytics • Programming

• Process Improvement

• Team Coordination & Leadership

• Debugging • MLOPS • LLM • RAG

• Fine-Tune • LLaM

TECHNICAL SKILLS

Languages: Python, Java, Android ,SQL, C, C++, Kotlin, JavaScript.

Database: MySQL, MongoDB, PostgreSQL

Big Data: Hadoop, Snowflake

Statistics: central tendency Measures, Hypothesis Test, Normal Distribution.

Machine Learning: Linear/Logistic Regression, Decision Tree, Random Forest Ensemble Trees, XGBoost, KNN

Time Series: Arima, Prophet, Neural Prophet.

Deep Learning: CNN, RNN, LSTM

NLP: Transformer, LLM, Gen AI, RAG, LangChain, VectorDatabase, LLaM

Vector Database: FAISS, Chroma

Cloud Computing: AWS, Azure, Firebase(GCP), AWS-Sagemaker.

PROFESSIONAL EXPERIENCE

Senior Machine Learning Developer

Nov 2023 - Present

Tech Mahindra

Bengaluru, IN

Mile Separation Claims Prediction and Mile Separation Claims GenAI Chatbot

- Successfully integrated machine learning models for binary classification, specifically designed to predict the approval or rejection of mile separation claims.
- Implemented GenAI using Large Language Models (LLMs) to enhance decision-making processes.
- Developed a chatbot to assist supervisors in making informed decisions by providing detailed responses about Paycode based on OneNote documents and agreements for specific pools or routes in the railroad industry.
- Created contextual embedding vectors to improve the chatbot's understanding and retrieval of relevant information, ensuring accurate and context-aware responses.
- Achieved significant reduction in response times and increased overall efficiency in managing mile separation claims, leading to more strategic and informed decision-making.
- Focused on reducing mistakes and minimizing manual approval of claims, thereby increasing profitability for customers and stakeholders.

Healthcare and Medical Leveraging LLM and Knowledge Base for Logistics & Supply Chain

- Successfully integrated LLM for NLU and data analysis, specifically tuned for healthcare logistics complexities.
- Implemented RAG strategies to enhance the retrieval of pertinent healthcare logistics data.
- Reduced response times and increased overall efficiency in managing healthcare supply chain disruptions.

Medicaid Category Recommendation Search Engine: Healthcare drug

- Developed healthcare drug recommendation system using Transformer encoder-decoder.
- Conducted data pre-processing and feature engineering for large-scale healthcare datasets.
- Fine-tuned Transformer model and improved model performance and Achieved positive user feedback for accuracy.
- Actively participated in knowledge-sharing with team and stayed updated on emerging technologies.

Senior Data Scientist

Jun 2022 - Oct 2023

Autodesk

Bengaluru, IN

Trend Analysis Forecast: Sale Forecasting

- worked on Exploratory Data Analysis (EDA) to understand historical sales trends and patterns.
- Utilized Neural Prophet model for time-series forecasting, accommodating for various factors influencing sales trends.
- Implemented Serverless Python framework to enable scalable and cost-efficient computing.
- Configured and deployed the solution on AWS, integrating services such as S3 for data storage and Lambda for serverless computation, Deployed Docker Image on ECR service.

Indirect Tax Forecast: Indirect Tax Forecasting, Tax Dashboard

- Developed forecasting models using advanced analytics to predict future indirect tax liabilities.
- Implemented time-series data to optimize input for the Neural Prophet model, enhancing forecasting accuracy.
- Collaborated with tax experts and financial stakeholders and incorporate changes into the forecasting model.
- Implemented a dynamic Tax Dashboard, facilitating real-time monitoring , decision-making for financial stakeholders.
- Integrated the Prophet model using the Serverless Framework for AWS Lambda functions.
- Configured AWS services, including S3 for data storage, Lambda for serverless compute, and API Gateway for endpoint exposure, Deployed Docker Image on ECR service.

Transaction & Entertainment Anomaly Detection: Fraud Transaction Anomaly Flag Normal Detection

- Developed and implemented machine learning models for the detection of fraudulent transactions and abnormal entertainment activities.
- Created multiple roles and approaches to detecting of Anomaly for fraud transaction.
- Utilized statistical methods & predictive modeling to enhance the accuracy & anomaly detection efficiency algorithms.
- Implemented a dynamic anomaly detection system that adapts to new patterns and continuously evolves to counter emerging threats.
- Successfully deployed the Anomaly Detection system, contributing to a 86% reduction in security incidents.

Libraries: Pandas, Numpy, PySpark, Scikit-learn, TensorsFlow, Pytorch, Serverless, Flask, OCR, OpenAI.

Tools: Tableau, Power BI, Kibana, MLFlow, ELK, Bamboo, CDSW, RabbitMQ.

Devops: Docker, Kubernetes, CI-CD Pipeline

ACHIEVEMENT & HACKATHON

- Conferred with the Stakeholder's **Best Award in May-2017** for exemplary performance in successfully completing and navigating a complex project, ensuring a seamless and successful delivery.
- Honored with the **Team Leadership and Knowledge Transfer Award** in **December 2021** for exemplary leadership in managing and guiding a high-performing team.
- **Banking ML Analytics:** 43th Rank
- **LTFS Fin Hack:** 57th Rank
- **Mobility Analytics:** 61th Rank
- **HR Analytics:** 82th Rank
- **Computer Vision:** 94th Rank

CERTIFICATIONS

- AWS Cloud Computing Certificate Jan 2022
- Machine Learning DataMites Certification June 2016
- Android Lara Certification June 2014

EDUCATION

MS in Data Science, Liverpool John Moore University, UK
Oct 2019 - Nov 2021

PG Diploma in Machine Learning & Artificial Intelligence, IIIT-Bangalore
Aug 2019 - Dec 2020

B.tech (ISE), Siddaganga Institute of Technology, Tumkur, Karnataka
Jun 2011 - Aug 2015

Sale Search Chatbot: Bot Conversation with Humans.

- Designed and implemented the chatbot's conversational flow, ensuring a seamless and intuitive interaction for users.
- Leveraged NLP techniques to enhance the chatbot's understanding of user queries and provide accurate responses.
- Integrated ML algorithms to continuously improve the chatbot's ability to understand and adapt to user preferences.
- Collaborated with the different team to identify common user queries and optimize the chatbot's responses.
- Enhanced the chatbot's capabilities to handle complex customers queries, resulting in increased customer engagement

AI Engg 2

Apr 2020 - Jun 2022

IQVIA

Bangalore, IN

Examination of Chest X-ray / CT-Scan COVID-19 Detection: Image Processing, Transfer Learning

- Preprocessed and curated a diverse dataset of Chest X-ray and CT-Scan images for model training and validation.
- Implemented transfer learning using pre-trained models to enhance the model's ability to detect COVID-19 patterns.
- Developed and fine-tuned a CNN model, optimizing hyper-parameters for maximum accuracy and sensitivity.
- Directed **model development**, validation, testing, and **implementation** of analytical products and applications
- Achieved a classification accuracy of 94.52 in detecting COVID-19 from medical images.

ETMF-PD-AI: Extract Document Details & Upload Document.

- Implementation of AI algorithms for accurate extraction of relevant document details.
- Development of a user-friendly interface for document upload, ensuring a seamless and intuitive experience.
- Coordinated with cross-functional teams to integrate the solution with existing document management systems.
- Implemented Docker containers to encapsulate micro-services, ensuring consistency across different environments.
- Utilized Kubernetes for container orchestration, enabling automatic scaling & efficient management of micro-services.
- Daily Checked and managed different virtual environments to accommodate diverse micro-services requirements.

GXP-TMS: Translation Management Service

- Worked on RabbitMQ for asynchronous communication between micro-services, enhancing system responsiveness.
- Established a robust version control system using GitLab, enhancing collaboration and code management.
- Worked on backend services using Python, incorporating AI algorithms for enhanced translation quality and efficiency.
- Successfully deployed GXP-TMS on Kubernetes, resulting in improved scalability and resource utilization.
- Designed and maintained the MySQL database for storing and retrieving translation data.

Data Science Consultant

Jul 2019 - Apr 2020

Gyansys Infotech

Bangalore, IN

Legal Contract : Extract Text and Predict Discount Money.

- Extracting multiple legal documents using OCR technique used Python Programming.
- Worked on Regular Expression to find pattern of Legal documents text.
- Worked on Data Cleaning, Data Pre-processing, Visualization used PandasProfile.
- Implemented Model Training, Model Compile, Model Evaluation used XGBoost.
- Worked Application & Model Deployment on Azure used Docker Images.

Senior Software Engineering

Feb 2018 - Jun 2019

High Peak Software Pvt Ltd

Bangalore, IN

LendStash: Predict Book and Not Book

- Implemented Recognize Text, Detecting different types of Faces, Scan Barcodes.
- Detect the book first page using OCR technique to fetch all text then storing text in string variables.
- Worked on Keras, TensorFlow, Image Augmentation, Image Resize Convolution, Relu Layer, Max-pooling Layer.
- Implemented Model building using book images dataset to perform image classification.
- Saving model using TFLite and keeping model on Google Firebase cloud used Python Programming.

Machine Learning Firebase Kit: App for Predicting All Types of Machine Learning on Mobile Device.

- Implemented Recognize Text, Detecting different types of Faces, Scan Barcodes.
- Worked on Label Images to fetch all dimension values, Firebase Vision Text.
- Added Text Recognition Processor, File, Detect Image, Firebase Vision Image used Kotlin & Java Programming.

Software Engineering

Nov 2015 - Feb 2018

Robosoft Technologies Pvt Ltd

Udupi, India

Wand: It is an enterprise application. It is a login based application designed for Manager and Worker.

- Worked on Login, Signup, Logout User Interface and Customised layout using different styles.
- Implemented Google Firebase for Database stores, Rest API used Java Programming, XML.
- Followed MVP design pattern and added unit testing for application validation.